

Digital Assignment 1

1 $F(A, B, C, D) = \sum (0, 1, 3, 8, 10, 11, 13)$

A

CD \ AB	00	01	11	10
00	1	1	1	1
01				
11		1		
10	1		1	1

$$= B'C'D' + A'B'D' + ABC'D + ABC$$

Alternate

$$= ABC'D + A'B'C + B'CD + AB'D'$$

2 $F(X, Y, Z) = \sum m(0, 1, 6, 7)$

BC \ A	00	01	11	10
0	1	1		
1			1	1

$$= A'B' + AB$$

3 $F(P, Q, R, S) = \prod M(1, 5, 6, 8, 11, 14)$
 $= \sum m(0, 2, 3, 4, 7, 9, 10, 12, 13, 15)$

RS \ PQ	00	01	11	10
00	1		1	1
01	1		1	
11		1		1
10	1	1	1	

$$= A'C'D' + B'C'D' + AC'D + AB'D + A'B'C + A'CD + ABCD$$

$$4 \quad F(A, B, C, D, E) = \sum m(0, 2, 4, 5, 8, 12, 13, 14, 20, 27, 28, 30, 31)$$

① Quine McCluskey Method

$F(A, B, C, D, E) = \sum m(0, 2, 4, 5, 8, 12, 13, 14, 20, 27, 28, 30, 31)$

	A	B	C	D	E
✓ 0	0	0	0	0	0
✓ 2	0	0	0	1	0
✓ 4	0	0	1	0	0
✓ 8	0	1	0	0	0
✓ 5	0	0	1	0	1
✓ 12	0	1	1	0	0
✓ 20	1	0	1	0	0
✓ 13	0	1	1	0	1
✓ 14	0	1	1	1	0
✓ 28	1	1	1	0	0
✓ 27	1	1	0	1	1
✓ 30	1	1	1	1	0
✓ 31	1	1	1	1	1

	A	B	C	D	E
0, 2	0	0	0	0	-
0, 4	0	0	1	0	-
0, 8	0	1	0	0	-
4, 5	0	0	1	0	1
4, 12	0	1	1	0	0
4, 20	1	0	1	0	0
8, 12	0	1	1	0	0
5, 13	0	1	1	0	1
12, 13	0	1	1	0	-
12, 14	0	1	1	1	0
20, 28	1	1	1	0	0
14, 30	1	1	1	1	0
28, 30	1	1	1	1	-
27, 31	1	1	1	1	1
30, 31	1	1	1	1	-

Prime Implicant Chart

	0, 2	4, 5	8, 12	13	14	20	27	28	30	31
$A'B'C'E'$	x									
$ABDE$										
$ABCD$										
$A'D'E$	x									
$A'CD$										
$CD'E'$										
BCE'										

$A'B'C'E' + ABDE + A'D'E + A'CD + CD'E' + BCE'$

$$5 \quad F(A, B, C, D, E) = \sum m(0, 1, 3, 7, 11, 13, 15)$$

	A	B	C	D	E
✓ 0	0	0	0	0	0
✓ 1	0	0	0	0	1
✓ 3	0	0	0	1	1
✓ 7	0	0	1	1	1
✓ 11	0	1	0	1	1
✓ 13	0	1	1	0	1
✓ 15	0	1	1	1	1

	A	B	C	D	E
0, 1	0	0	0	0	-
1, 3	0	0	0	1	-
3, 7	0	0	1	1	-
3, 11	0	1	0	1	1
7, 15	0	1	1	1	1
7, 13	0	1	1	0	1

	0	1	3	7	11	13	15
$\bar{A}\bar{B}\bar{C}\bar{D}$	x						
$\bar{A}\bar{B}C\bar{E}$							
$\bar{A}DE$							
$A'BCE$							

$$= \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}DE + \bar{A}BCE$$

Q.2

i) $(11011.0101)_2 = (9)_{10} = (9)_8 = (9)_{16}$

$16 \quad 8 \quad 4 \quad 2 \quad 1$

$1 \quad 1 \quad 0 \quad 1 \quad 1$

$16+8 \quad +2+1$

$= 25$

$0101 = \frac{1}{2} + \frac{1}{16} = \frac{5}{16} = 0.3125$

$= (25.3125)_{10}$

$011 \quad 011 \quad . \quad 010 \quad 100$

$(33.24)_8$

A	B	C	D	E	F
10	11	12	13	14	15

$\underline{00011011} \quad . \quad \underline{0101}$
 $(18.5)_{16}$

ii) $(CA75)_{16} = (9)_{10} = (9)_8 = (9)_2$

CA75

$(1100100101110101)_2$
 $(144565)_8$

$2^0 + 2^2 + 2^4 + 2^5 + 2^6 + 2^8 + 2^{11} + 2^{14} + 2^{15}$
 $1+2+16+32+64+256+2048+16384+32768$

$= 51573$

iii) $(9741.12)_{10} = (?)_8$

9	7	4	1	
1	4	8	7	0
0	2	4	3	5
1	1	2	1	7
1		6	0	8
0		3	0	4
0		1	5	2
0		7	6	
0		3	8	
0		1	9	
1		9		
1		4		

$= (10110000011001.0001)_2$

$= (26031.04)_8$

	0	1	2			
0		0	2	4		
0			0	4	8	
0				1	9	6

$$iv) (107)_8 = (9)_{16}$$

$$\underline{001000111} = (47)_4$$

$$v) (56088/123)_{10} = (710)_9$$

$$\frac{56088}{123} = (456)_{10}$$

	456	
0	228	
0	114	
0	57	
1	28	
0	14	
0	7	
1	3	
1	1	

$$\frac{111001000}{(710)_8}$$

$$vi) (ABDA)_{16} + (3AC9)_{16}$$

$$= \begin{array}{r} \begin{array}{cccccccc} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 \end{array} + \\ \begin{array}{cccccccc} 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 1 \end{array} \\ \hline \begin{array}{cccccccc} 1 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \end{array} \\ \hline \end{array}$$

$$(E6A3)_{16}$$

$$vii) (7531)_8 - (4442)_8$$

$$2' \text{ complement } \rightarrow (4442)_8 \rightarrow (100100100010)_2$$

$$\rightarrow 01101011101 + 1$$

$$110101110$$

$$(7531)_8 \rightarrow \begin{array}{r} \begin{array}{cccccccc} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{array} + \\ \begin{array}{cccccccc} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 1 & 1 & 0 \end{array} \\ \hline \begin{array}{cccccccc} 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 & 1 \end{array} \\ \hline \end{array} \rightarrow (3067)_8$$

VIII $(0110101)_2 - (1101011)_2$

Z' complement

	1	1	1	1
	0	0	1	0
	1	0	1	0
	0	1	1	0
	1	0	0	1

→ $(1001010)_2$ 2's complement
- $(0110110)_2$

$$1x \quad (EDCA)_{16} - (f95A)_4$$

Smaller - bigger = - Answer
 BDC A - (-2' bigger) = (Answer) 2' complement

$$(f_{95A})_{16} = (11 \ 111001010101010)_2$$

EDCA =

$$\begin{array}{r} 2^{\text{Com}} = 000001101010110 \\ 1110110111001010 \\ \hline (1111010001110000) \quad 2^{\text{Complement}} \\ \hline 0000101110010000 \\ (-B90)_{16} \end{array}$$

x) $(1101011)_2 - (10111)_2$

$$\begin{pmatrix} 2 \text{ Com} & 0010111 \\ & (110100) \end{pmatrix}_2$$

$$\begin{array}{r} 1111 \\ 1101011 \\ 1101001 \\ \hline \times 1010100 \\ (1010100)_2 \end{array}$$

Q3

Binary to Gray